METRIC

MIL-PRF-32701
3 March 2022

PERFORMANCE SPECIFICATION
BRAKE FLUID, POLYGLYCOL, AUTOMOTIVE, ALL-WEATHER, OPERATIONAL AND PRESERVATIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers polyglycol-based hydraulic brake fluid for use in hydraulic brake systems at ambient temperatures ranging from -55°C to 55 °C and operational temperatures down to -40 °C (see 6.1). This fluid is primarily intended for use in fully hydraulic (i.e. not actuated by air) anti-lock braking systems (ABS). This fluid is incompatible with silicone- based brake fluid as specified in MIL-PRF-46176. The fluid is identified by military symbol PBF.

Comments, suggestions, or questions on this document should be addressed to U.S. Army Ground Vehicle Systems Center, ATTN: FCDD-GVS-SAT MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000, or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9150

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of the list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E29	-	Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications
ASTM E3244	-	Standard Practice for Integrity Assurance and Testing of Single-Use Systems

(Copies of these documents are available online at www.astm.org).

SAE INTERNATIONAL

SAE J1704	-	Motor Vehicle Brake Fluid Based Upon Glycols, Glycol Ethers, and the Corresponding Borates

(Copies of this document is available online at https://www.sae.org/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Polyglycol brake fluids furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified product list before contract award (see 4.1.1 and 6.2). Any change in the formulation of a qualified product will necessitate its requalification.

3.2 Design, materials, and manufacturing processes. Unless otherwise specified, the design, materials, and manufacturing process selection is the prerogative of the contractor as long as all articles submitted to the government fully meet the requirements specified herein.

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3.3 Performance Requirements. The properties of the brake fluid shall be as specified in table I and 3.3.1 through 3.3.11.

TABLE I. Brake fluid properties and test methods.

Property	Test Limits	Test Method
Equilibrium reflux boiling point (ERBP), °C, min (dry)	260	SAE J1704 Equilibrium Reflux Boiling Point
Wet ERBP, °C, min	170	SAE J1704 Wet Equilibrium Reflux Boiling Point
Kinematic Viscosity, @ 100 °C, mm^2/s, min	1.5	SAE J1704 Viscosity
Kinematic Viscosity, @ -40 °C, mm^2/s, max	900	SAE J1704 Viscosity
pH value	7.0 to 11.5	SAE J1704 pH Value
Brake Fluid Stability		
a) High Temperature Stability, @ 185±2 °C for 2 hrs	See 3.3.4.1	SAE J1704 High Temperature Stability
b) Chemical Stability	See 3.3.4.2	SAE J1704 Chemical Stability
Corrosion (wet), @ 100±2 °C for 120±2 hrs on SBR & EPDM	See 3.3.5	SAE J1704 Wet Corrosion Test[1]
a) Metal test strip weight change, mg/cm^2, max		
1) Steel, tinned iron, cast iron	0.1	
2) Aluminum	0.1	
3) Brass, Copper	0.2	
b) Condition of metal strips	See 3.3.5.2	
c) Condition of brake fluid @ 25±5 °C	See 3.3.5.3	
d) Sedimentation, vol%, max	0.10	
e) pH value	7.0 – 11.5	
f) Appearance of cups	See 3.3.5.6	
g) Cup hardness decrease, IRHD, max		
1) SBR	15	
2) EPDM	10	
h) Base diameter increase, SBR only, mm, max	1.4	
i) Volume change, EPDM only, vol%, max	10	
Low Temperature Stability @ -55±2 °C for 6±0.2 hrs		SAE J1704 Fluidity and Appearance at Low Temperature At -50 °C (-58 °F)
a) Stratification, sedimentation, crystallization	None	
b) Bubble flow time, seconds, max	35	
c) Appearance @ 23±5 °C	Original clarity	

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TABLE I. Brake fluid properties and test methods – Continued.

Property	Test Limits	Test Method
Water Tolerance (Tolerance to High Humidity)		SAE J1704 Water Tolerance
a) Low Temperature @ -40±2 °C for 22±2 hrs		SAE J1704 Water Tolerance At -40 °C (-40 °F)
1) Stratification, separation, precipitation, crystallization	None	
2) Appearance	See 3.3.7.1	
3) Bubble flow time, seconds, max	10	
b) Appearance @ 25±5 °C for 4±0.5 hrs	Original clarity	See 4.3.10.2
c) High Temperature @ 60±2 °C for 22±2 hrs		SAE J1704 Water Tolerance At 60 °C (140 °F)
1) Stratification	None	
2) Clarity	Clear	
3) Sedimentation, vol%, max	0.05	
Compatibility		SAE J1704 Compatibility
a) Low Temperature @ -40±2 °C for 22±2 hrs		SAE J1704 Compatibility At -40 °C (-40 °F)
1) Stratification, sedimentation, crystallization	None	
2) Appearance	See 3.3.8.1	
b) High Temperature @ 60±2 °C for 22±2 hrs		SAE J1704 Compatibility At 60 °C (140 °F)
1) Sedimentation, vol%, max	0.05	
2) Stratification	None	
3) Clarity	Clear	
Resistance to Oxidation		SAE J1704 Resistance to Oxidation
a) Appearance	See 3.3.9.1	
b) Gum	See 3.3.9.2	
c) Metal test strip weight change, mg/cm^2, max		
1) Aluminum	0.05	
2) Cast iron	0.3	
Effects on Rubber @ 120±2 °C for 70±2 hrs, SBR		SAE J1704 Effect on Rubber
a) Base diameter increase, mm, max	0.15 to 1.40	
b) Hardness decrease, IRHD, max	15	
c) Appearance	See 3.3.10.3	
Fluid Color	Colorless to Amber	N/A

NOTES:
1. Wet corrosion test is performed with 5% water.

3.3.1 ERBP and wet ERBP. The brake fluid shall meet the requirements shown in table I for ERBP and wet ERBP. See 4.3.2 and 4.3.3 for the verification test methods.

3.3.2 Kinematic viscosity. The brake fluid shall meet the requirements shown in table I for kinematic viscosity at 100 and -40 °C. See 4.3.4 for the verification test method.

3.3.3 pH value. The brake fluid shall meet the requirements shown in table I for the pH value. See 4.3.5 for the verification test method.

3.3.4 Brake fluid stability. The brake fluid shall meet the requirements in 3.3.4.1 and 3.3.4.2. See 4.3.6 and 4.3.7 for the verification test methods.

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3.3.4.1 High temperature stability. The brake fluid shall not have its ERBP decrease by more than 5 °C from the ERBP determined in 3.3.1 prior to exposure at high temperature.

3.3.4.2 Chemical stability. The brake fluid shall not have its ERBP change by more than 5 °C from the maximum fluid temperature recorded in the verification test method in 4.3.7.

3.3.5 Corrosion (wet). The brake fluid shall meet the requirements shown in table I and in 3.3.5.1 through 3.3.5.9. See 4.3.8 for the verification test method.

3.3.5.1 Metal test strip weight change. The permissible change in weight of the strips shall be as specified in table I after exposure to the brake fluid as specified in 4.3.8.

3.3.5.2 Condition of metal strips. The metal strips shall not exhibit pitting or roughness outside contact area after completion of the corrosion test as specified in 4.3.8. Staining or discoloration of the metal test strips is permitted.

3.3.5.3 Condition of brake fluid. The brake fluid shall exhibit no gelling at 25±5 °C, and no crystallization deposit shall form after completion of the corrosion test as specified in 4.3.8.

3.3.5.4 Sedimentation. The brake fluid shall meet the sedimentation requirement shown in table I after completion of the corrosion test as specified in 4.3.8.

3.3.5.5 pH value. The brake fluid shall meet the pH requirements shown in table I after completion of the corrosion test as specified in 4.3.8.

3.3.5.6 Appearance of cups. The rubber cups shall not show any signs of sloughing, tackiness, blisters, or any other form of disintegration after completion of the corrosion test as specified in 4.3.8.

3.3.5.7 Cup hardness decrease. The decrease in the hardness of any rubber cup shall meet the requirements shown in table I after completion of the corrosion test as specified in 4.3.8.

3.3.5.8 Base diameter increase. The increase in the base diameter of the styrene-butadiene rubber (SBR) cup shall meet the requirements shown in table I after completion of the corrosion test as specified in 4.3.8.

3.3.5.9 Volume change. The change in volume of the ethylene propylene diene monomer (EPDM) rubber cup shall meet the requirements shown in table I after completion of the corrosion test as specified in 4.3.8.

3.3.6 Low temperature stability. The brake fluid shall meet the low temperature stability requirements shown in table I and in 3.3.6.1 through 3.3.6.3. See 4.3.9 for the verification test method.

3.3.6.1 Stratification, sedimentation, crystallization. The brake fluid shall meet the requirements shown in table I.

3.3.6.2 Bubble flow time. The brake fluid shall meet the bubble flow time requirements

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shown in table I.

3.3.6.3 Appearance @ 23 ± 3°C. The brake fluid shall meet the appearance requirements shown in table I.

3.3.7 Water tolerance. The brake fluid shall meet the water tolerance requirements shown in table I and in 3.3.7.1 through 3.3.7.3. See 4.3.10 for the verification test method.

3.3.7.1 At low temperature. The brake fluid shall meet the requirements shown in table I. The contrast lines on a hiding power chart[1] shall be discernable when viewed through the brake fluid.

3.3.7.2 Appearance at 25 °C. The brake fluid shall meet the requirements shown in table I.

3.3.7.3 At high temperature. The brake fluid shall meet the requirements shown in table I.

3.3.8 Compatibility. The brake fluid shall meet the compatibility requirements shown in table I and in 3.3.8.1. See 4.3.11 for the verification test method.

3.3.8.1 At low temperature. The brake fluid shall meet the requirements shown in table I. The contrast lines on a hiding power chart shall be discernable when viewed through the brake fluid.

3.3.8.2 At high temperature. The brake fluid shall meet the requirements shown in table I.

3.3.9 Resistance to oxidation. The brake fluid shall meet the oxidation resistance requirements shown in table I and in 3.3.9.1 through 3.3.9.3. See 4.3.12 for the verification test method.

3.3.9.1 Appearance. The metal test strips shall exhibit no pitting or roughness outside contact area; staining and discoloration is permitted.

3.3.9.2 Gum. The metal test strips shall exhibit no more than a trace of gum.

3.3.9.3 Weight change. The metal test strips shall meet the weight change requirements shown in table I.

3.3.10 Effect on rubber. The brake fluid shall meet the effect on rubber requirementsshown in table I and in 3.3.10.3. See 4.3.13 for the verification test method.

3.3.10.1 Base diameter increase. The rubber cups shall meet the base diameter increase requirements shown in table I.

[1] A suitable hiding power chart is described in SAE J1704 as RM-28, published by SAE International.

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3.3.10.2 Hardness decrease. The rubber cups shall meet the hardness decrease requirements shown in table I.

3.3.10.3 Appearance. The rubber cups shall show no sloughing, tackiness, blisters, or any other form of disintegration after exposure to the brake fluid.

3.3.11 Fluid color. The brake fluid shall meet the color requirements lists in table I.

3.3.12 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

4.1.1 Qualification inspections. Qualification inspections (See 4.2).

4.2 Qualification inspections. Qualification inspections shall consist of all verifications listed in table I and 4.3.

4.3 Verification test procedures.

4.3.1 Testing. Tests shall be conducted as specified in table I and 4.3.2 through 4.3.13. The testing specified in this section includes only the most recently developed category tests. For purposes of determining conformance with each requirement, an observed value or calculated value shall be rounded off to the nearest unit in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off procedure given in ASTM E29. All values, unless otherwise stated or agreed upon by the qualifying activity are critical specifications as defined by ASTM E3244.

4.3.2 Equilibrium reflux boiling point. The equilibrium reflux boiling point of the brake fluid, as required in 3.3.1, shall be determined in accordance with the Equilibrium Reflux Boiling Point section of SAE J1704.

4.3.3 Wet boiling point. The wet boiling point of the brake fluid, as required in 3.3.1, shall be determined in accordance with the Wet Equilibrium Reflux Boiling Point section of SAE J1704.

4.3.4 Kinematic viscosity. The kinematic viscosity of the brake fluid at -40 and 100 °C, as required in 3.3.2, shall be determined in accordance with the Viscosity section of SAE J1704.

4.3.5 pH value. The pH value of the brake fluid, as required by 3.3.3, shall be determined in accordance with the pH Value section of SAE J1704.

4.3.6 High temperature stability. The high temperature stability of the brake fluid, as required by 3.3.4.1, shall be determined in accordance with the High Temperature Stability section

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of SAE J1704.

4.3.7 Chemical stability. The chemical stability of the brake fluid, as required by 3.3.4.2, shall be determined in accordance with the Chemical Stability section of SAE J1704.

4.3.8 Corrosion (wet). The corrosion resistance of the brake fluid, as required by 3.3.5, shall be determined in accordance with the Wet Corrosion section of SAE J1704. Only test the metal specimens and rubber cups designated in table I of this document.

4.3.9 Low temperature stability. The low temperature stability of the brake fluid, as required by 3.3.6, shall be determined at -55±2 °C for 6±0.2 hours in accordance with the "at -50 °C (-58 °F)" subsection of the Fluidity and Appearance at Low Temperature section of SAEJ1704.

4.3.10 Water tolerance. The water tolerance of the brake fluid, as required by 3.3.7, shall be determined in accordance with the Water Tolerance section of SAE J1704.

4.3.10.1 At low temperature. The low temperature water tolerance of the brake fluid, as required by 3.3.7.1, shall be determined in accordance with the "at -40 °C (-40 °F)" subsection of the Water Tolerance section of SAE J1704.

4.3.10.2 Appearance at 25 °C. Following the verification specified in 4.3.10.1, allow the brake fluid to stabilize at 25±5 °C for 4±0.5 hours. Visually compare the test fluid to an as-received sample of the test fluid in an identical container to verify conformance to the requirement in 3.3.7.2.

4.3.10.3 At high temperature. The high temperature water tolerance of the brake fluid, as required by 3.3.7.3, shall be determined in accordance with the "at 60 °C (140 °F)" subsection of the Water Tolerance section of SAE J1704.

4.3.11 Compatibility. The compatibility of the brake fluid, as required by 3.3.8, shall be determined in accordance with the Compatibility section of SAE J1704.

4.3.12 Resistance to oxidation. The oxidation resistance of the brake fluid, as requiredby 3.3.9, shall be determined in accordance with the Resistance to Oxidation section of SAE J1704.

4.3.13 Effect on rubber. The effect on cups of the brake fluid, as required by 3.3.10, shall be determined in accordance with the Effect on Rubber section of SAE J1704.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military

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Department's or Defense Agency's automated packaging files, CD-ROM products, or bycontacting the responsible packaging activity.

5.1.1 Labeling. Packaging shall be labeled using the military symbol (see 1.1).

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory).

6.1 Intended use. The brake fluid covered by this specification is military unique due to the more stringent requirements in the areas of boiling point (both wet and dry), low temperature viscosity, and corrosion protection. It is intended for use as an operational fluid and preservative fluid in automotive fully hydraulic (i.e. not actuated by air) anti-lock brake systems (ABS) at ambient temperatures ranging from -55 °C to 55 °C; the brake fluid itself is only operational down to fluid temperatures of -40 °C. While compatible, this fluid is not intended to be used in military ground equipment without ABS, or in systems that utilize an air-over-hydraulic ABS where the brake fluid is not usedto actuate the ABS. In these cases, the standard military silicone brake fluid under MIL-PRF-46176should be used. Table II lists the National Stock Numbers (NSN) that have been established for MIL-PRF-XXXXX.

TABLE II. National Stock Numbers (NSNs) for PBF.

NSN	Unit of Issue
9150-01-694-1600	1-quart
9150-01-671-8502	55-gallon

As a general recommendation, the drain interval on PBF is two years. At this time, use of PBF should be approved by a Lubrication Order, Technical Manual, Maintenance Action/Information Message, or by a representative of the equipment's Product Manager (PM) or other designated representative (e.g., equipment specialist/LAR).

6.1.1 Replacement of brake fluid. If brake fluid in accordance with this specification is used to replace another type of polyglycol brake fluid (e.g. DOT 3, DOT 4), adequate flushing of the brake system must be accomplished to remove all traces of the brake fluid that is to be replaced. If these fluids are not completely removed, the viscosity properties and high boiling point protection can be somewhat diminished. If changing from a silicone brake fluid (e.g. DOT 5, MIL-PRF-46176), contact U.S. Army DEVCOM GVSC for guidance and instructions; silicone brake fluids have a different base chemistry and are not miscible with water unlike polyglycol brake fluids. This incompatibility will cause phase separation and possible damage to the brake system if not properly flushed.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Qualification (see 3.1).

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c. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 32701 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from:

U.S. Army DEVCOM GVSC
6501 E. 11 Mile Road
ATTN: Fuels & Lubricants Branch
FCDD-GVS-ES (MS 110)
Detroit Arsenal, MI 48397-5000.
usarmy.detroit.rdecom.mbx.tardec-pol-help@army.mil

Products will not be considered for inclusion in QPL-32701 until such time as appropriate departmental medical activity has reviewed all pertinent safety data sheets (FED-STD-313).

6.4 Test materials. Rubber cups, metal strips, hiding power charts, corrosiveness test jars, and the SAE compatibility fluid can be obtained from the SAE International, 400 Commonwealth Drive, Warrendale, PA 15096.

6.5 Interchangeability. Brake fluid covered by this specification cannot be used in the same equipment and systems as the brake fluids covered by MIL-PRF-46176.

6.6 Safety Data Sheets (SDS). Contracting officers should identify those activities requiring copies of SDS's prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313; and 29 CFR 1910.1200 requires thatthe SDS for each hazardous chemical used in an operation must be readily available to personnelusing the material. Contracting officers should identify the activities requiring copies of the SDS.

6.7 Simulated Service. A simulated service test rig built to evaluate the compatibility of a fluid under heavy ABS conditions has been developed; a test method will be added to a later revision of this specification as a requirement for all future qualification efforts.

6.8 Subject term (key word) listing.

Hydraulic
NATO STANAG 1135
Viscosity

6.9 International standardization agreement implementation. This specification implements STANAG 1135, "Interchangeability of Fuels, Lubricants and Associated Products

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Preparing Activity:
Army – AT
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Review Activities:
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